INCH-POUND

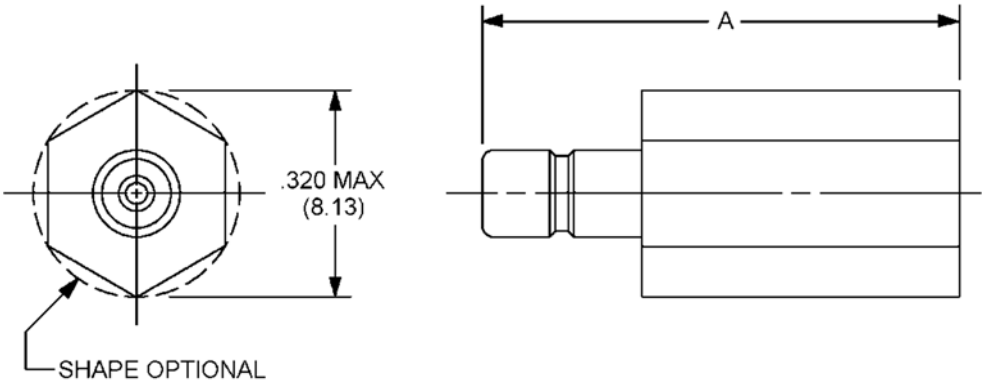
MIL-PRF-39012/68D
w/AMENDMENT 1
10 August 2016
SUPERSEDING
MIL-PRF-39012/68D
16 November 2011

PERFORMANCE SPECIFICATION SHEET

# CONNECTORS, PLUGS, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMB (CABLED, PIN CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39012.




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for information only.
4. For dimension A, see table I.
5. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28, appendix 10.
6. Dimension A defines the overall length of the connector when assembled to the cable.
7. Maximum overall diameter of the connector is .320 inch (8.13 mm).
8. All undimensioned pictorial configurations are for reference purposes only.
9. Series SMB, pin contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers, cross-reference, and dimensions.

| Dash No.<br><br>M39012/68- | # Applicable cable group from MIL-PRF-39012, appendix B<br>M17/ | Typical mating Connector<br>1/ | Dimensions | Inches (millimeters) Maximum<br>2/ 5/ |
|---|---|---|---|---|
| CATEGORY A – FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED) | | | | |
| 0003 3/ 4/ | Cable group I<br><br>93-RG178 * @<br>169-00001 % | M39012/67-0003<br>M39012/69-0003 | A | .88 (22.35) |
| 0004 3/ 4/ | Cable group II<br><br>113-RG316 * @<br>172-00001 %<br>94-RG179 ^ | M39012/67-0004<br>M39012/69-0004 | | |
| 0103 3/ | Cable group I<br><br>93-RG178 * @<br>169-00001 % | M39012/67-0103<br>M39012/69-0103 | | |
| 0104 3/ | Cable group II<br><br>113-RG316 * @<br>172-00001 %<br>94-RG179 ^ | M39012/67-0104<br>M39012/69-0104 | | |
| 4101 3/ | Cable group I<br><br>93-RG178 * @<br>169-00001 % | M39012/67-4101<br>M39012/69-4101 | | |
| 4102 3/ | Cable group II<br><br>113-RG316 * @<br>172-00001 %<br>94-RG179 ^ | M39012/67-4102<br>M39012/69-4102 | | |

See notes at end of table.



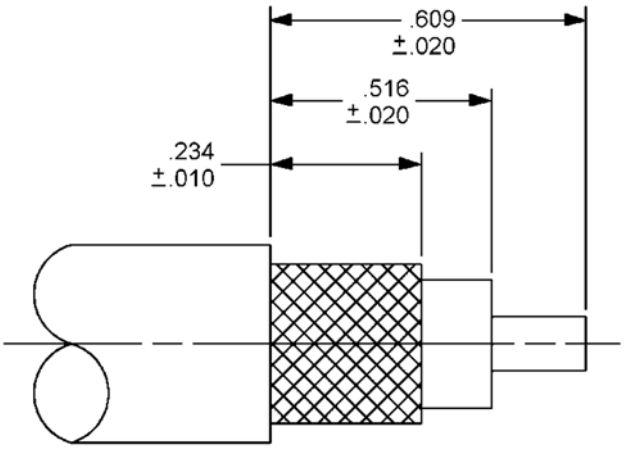
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers, cross-reference, and dimensions – Continued.

| Dash No.<br><br>M39012/68- | # Applicable cable group from MIL-PRF-39012, appendix B<br>M17/ | Typical mating connector<br>1/ | Dimensions | Inches 2/<br>(millimeters)<br>Maximum |
|---|---|---|---|---|
| CATEGORY C – FIELD REPLACEABLE (MIL-C-22520/5 BASIC CRIMP TOOL) 5/ | | | | |
| 0011 3/ 4/ | Cable group I ~<br><br>93-RG178 * @<br>169-00001 % | M39012/67-0011<br>M39012/69-0011 | A | 1.10 (27.94) |
| 0012 3/ 4/ | Cable group IIa &<br><br>113-RG316 * @<br>172-00001 %<br>94-RG179 ^ | M39012/67-0012<br>M39012/69-0012 | | |
| 0016 3/ | Cable group I ~<br><br>93-RG178 * @<br>169-00001 % | M39012/67-0016<br>M39012/69-0016 | | |
| 0017 3/ | Cable group IIa &<br><br>113-RG316 * @<br>172-00001 %<br>94-RG179 ^ | M39012/67-0017<br>M39012/69-0017 | | |
| 4001 3/ | Cable group I ~<br><br>93-RG178 * @<br>169-00001 % | M39012/67-4001<br>M39012/69-4001 | | |
| 4002 3/ | Cable group IIa &<br><br>113-RG316 * @<br>172-00001 %<br>94-RG179 ^ | M39012/67-4002<br>M39012/69-4002 | | |

1/ Mating connector Part or Identifying Number (PIN) is for reference only. All mating connectors may not be listed.
2/ Dimensions are in inches. Metric equivalents are given for information only.
3/ These connectors have captivated center contacts.
4/ Connector bodies shall be gold plated in accordance with ASTM B488, type II, code C, class 1.27.
5/ Category C connectors are assembled by means of the applicable crimping tool in accordance with MIL-DTL-22520 to the specified cable stripped as specified on figure 2.
# The latest version of each cable shall be applicable.
* Cable to be used when performing tests requiring cable except as in notes @ and ^.
@ Cable to be used for the +200°C thermal shock tests.
^ These are not 50 ohm cables, therefore, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.
~ M22520/5-33 closure B or M22520/5-03 closure B.
& M22520/5-35 closure B or M22520/503 closure A.
% Caution is directed to the applicable of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.


Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .010 | 0.25 |
| .020 | 0.51 |
| .234 | 5.94 |
| .516 | 13.11 |
| .609 | 15.47 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 2. Cable stripping dimensions for category C connectors.

ENGINEERING DATA:

Nominal impedance: 50 ohms.
Frequency range: 0 to 4 GHz.
Voltage rating: The voltage rating shall be as specified in table II.

TABLE II. Voltage rating.

| Cables | Voltage max. (sea level) V rms | Voltage max. at 70,000 ft. (4.437 kPa) V rms |
|---|---|---|
| Cable group I | 250 | 60 |
| Cable group II, IIa, | 335 | 85 |

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.
Force to engage and disengage:
Longitudinal force: Not applicable.
Torque: Not applicable.
Coupling proof torque: Not applicable.
Inspection conditions: Not applicable.
Mating characteristics: See MIL-STD-348.
Hermetic seal: Not applicable.
Leakage (pressurized connectors): Not applicable.


Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.

Insulation resistance: In accordance with MIL-STD-202-302, test condition B, 1,000 megohms minimum.

Center contact retention:

Axial force: 4 pounds (17.79 N), minimum axial force. Applicable to captivated center contacts only.

Radial torque: Not applicable.

Corrosion (salt spray): In accordance with MIL-STD-202-101, test condition B.

Voltage standing wave ratio (VSWR): From 500 MHz to 4 GHz, or approximately 80 percent of the upper cutoff frequency of the cable, whichever is lower.

| Cable group | VSWR |
|---|---|
| I | 1.30 + 0.04 (F) GHz |
| II, IIa, | 1.25 + 0.04 (F) GHz |

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.05 +.0025F (F in GHz).

Item 16: VSWR shall be less than 1.05 +.0025F (F in GHz).

Second step of VSWR checkout procedure – VSWR shall be less than 1.10 +.01F (F in GHz).

Group B inspection: VSWR shall be less than 1.08 +.017F (F in GHz).

Qualification and group C inspection: VSWR shall not exceed 1.15.

Connector durability: 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet mating characteristics and force to engage and disengage requirements.

Initial: 14 pounds (62.27 N), maximum.

Final: 14 pounds (62.27 N), maximum engage and disengage; 2 pounds (8.90 N) minimum disengagement.

Contact resistance: In milliohms, maximum.

| | Initial | After environment |
|---|---|---|
| Center contact: | 6.0 | 8.0 |
| Outer contact: | 1.0 | 1.5 |
| Braid to body: | 1 | Not applicable |

Dielectric withstanding voltage at sea level: In accordance with MIL-STD-202-301.

| Cable group | V rms (at sea level) |
|---|---|
| I | 750 |
| II, IIa | 1,000 |

Vibration, high frequency: In accordance with MIL-STD-202-204, test condition B.

Shock: In accordance with MIL-STD-202-213, test condition B.

Thermal shock: In accordance with MIL-STD-202-107, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and III).



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.

Moisture resistance: Not applicable.

Corona level:

Altitude: 70,000 feet (4.437 kPa).

| Cable group | Volts, min. |
|---|---|
| I | 185 |
| II, IIa | 250 |

RF high potential withstanding voltage:
Voltage and frequency: Tested at a frequency from 5 to 7.5 MHz.
Leakage current: Not applicable.

| Cable group | Volts, rms min. |
|---|---|
| I | 500 |
| II, IIa | 700 |

Cable retention force: The cable retention force shall be as specified in table III.

TABLE III. Cable retention force. 1/ 2/

| Cable dielectric Outer diameter | Pounds, min. (N) | |
|---|---|---|
| | Non-crimp Single braid | Crimp Single braid |
| | Center contact | Center contact |
| | Captive | Captive |
| Inches, max. (mm) | | |
| .036 (0.91) | 13 (57.83) | 13 (57.83) |
| .066 (1.68) | 25 (111.20) | 25 (111.20) |

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

Coupling mechanism retention force: Not applicable.
RF leakage: -55 dB minimum, tested at frequency between 2 and 3 GHz..
RF insertion loss: .30 dB maximum tested at 4 GHz.

PIN: M39012/68 (dash number in table I or IV as applicable).

Group qualification: See table V.

Cross reference of PIN: See table VI.

Maintenance replacements for category B: See table VII.


Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.

TABLE IV. Category B – non-field replaceable (special tools may be required).

NOT FOR ARMY, NAVY, OR AIR FORCE USE. FOR OEM USE ONLY

| Dash No. 1/<br><br>M39012/68 | # Applicable cable group from MIL-PRF-39012, appendix B M17/ | Typical mating connector 2/ | Dimensions | Inches (millimeters) maximum 5/ 6/ |
|---|---|---|---|---|
| B0008 3/ 4/ | 93-RG178 * @<br>169-00001 % | M39012/67B0008<br>M39012/69B0008 | A | 1.10 (27.94) |
| B0009 3/ 4/ | 113-RG316 * @<br>172-00001 %<br>119-RG174<br>173-00001 % | M39012/67B0009<br>M39012/69B0009 | | |
| B0010 3/ 4/ | 94-RG179 * @ ^ | M39012/67B0010<br>M39012/69B0010 | | |
| B0013 3/ | 93-RG178 * @<br>169-00001 % | M39012/67B0008<br>M39012/69B0008 | | |
| B0014 3/ | 113-RG316 * @<br>172-00001 %<br>119-RG174<br>173-00001 % | M39012/67B0009<br>M39012/69B0009 | | |
| B0015 3/ | 94-RG179 ^ @ * | M39012/67B0010<br>M39012/69B0010 | | |

1/ For cross-reference of dash number to superseded PIN or type designation, see table VI.
2/ Mating connector PIN are for reference only. All mating connectors may not be listed.
3/ These connectors have captive center contacts.
4/ Connector bodies shall be gold plated in accordance with ASTM B488, type II, code C, class 1.27.
5/ Dimensions are in inches.
6/ Metric equivalents are given for information only.
# The latest version of each cable shall be applicable.
* Cable to be used when performing tests requiring cable except as in note @.
@ Cable to be used for the +200°C thermal shock tests.
% Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.
^ These are not 50 ohm cables, therefore, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.

NOTE: Connectors mate with connectors of the same material only; Example: M39012/68-3001 mates with M39012/67-3001, and M39012/68-4001 mates with M39012/67-4001.


Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.

TABLE V. Group qualification and retention testing.

| Group | Submission and retention of Qualification of any of the following connectors 1/ M39012/ | Qualifies the following connectors M39012/ |
|---|---|---|
| I | 68-0003<br>70-0003 | 68-0003<br>68-0103 2/<br>70-0003<br>70-0103 2/ |
| II | 68-0004<br>70-0004 | 68-0004<br>68-0104 2/<br>70-0004<br>70-0104 2/ |
| III | 68B0008<br>70B0008 | 68B0008<br>68B0013 2/<br>70B0008<br>70B0013 2/ |
| IV | 68B0009<br>70B0009 | 68B0009<br>68B0010<br>68B0014 2/<br>68B0015 2/<br>70B0009<br>70B0010<br>70B0014 2/<br>70B0015 2/ |
| V | 68B0010<br>70B0010 | 68B0010<br>68B0015<br>70B0010<br>70B0015 2/ |
| VI | 68-0011<br>70-0011 | 68-0011<br>68-0016<br>70-0011<br>70-0016 2/ |
| VII | 68-0012<br>70-0012 | 68-0012<br>68-0017<br>70-0012<br>70-0017 2/ |

1/ Individual connectors other than listed are self qualifying only.
2/ Corrosion and contact resistance test data must be submitted to DSCC-VQ before qualification approval may be granted.

NOTES:

1. If a connector manufacturer produces a connector which meets all the requirements for two or more connector PINs (within the same series), the manufacturer may receive qualification approval for two or more connector PINs by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate PIN. For group qualification, the connectors must be of similar design.
2. For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right-hand column. The part does not necessarily have to be the part initially qualified.


Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.

TABLE VI. Cross reference of PIN.

| Preferred PIN M39012/68 | Substitute for PIN Or type designation |
|---|---|
| -0003 | M39012/68-0001 |
| -0004 | M39012/68-0002 |
| B0008 | M39012/68-0008, M39012/68-0005 |
| B0009 | M39012/68-0009, M39012/68-0006 |
| B0010 | M39012/68-0010, M39012/68-0007 |

TABLE VII. Maintenance replacements for category B.

| Category B number 1/ | Category C dash number | Category A dash number |
|---|---|---|
| B0008 | 0011 | 0003 |
| B0009 | 0012 | 0004 |
| B0010 | 0012 | 0004 |
| B0013 | 0016 | 0103 |
| B0014 | 0017 | 0104 |
| B0015 | 0017 | 0104 |

1/ Category B connectors are for original installation only. They will not be stocked or procured by the government.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-39012, this document references the following:

| | | |
|---|---|---|
| FED-STD-H28 | MIL-STD-202-204 | MIL-STD-348 |
| MIL-STD-202-101 | MIL-STD-202-213 | MIL-DTL-22520 |
| MIL-STD-202-107 | MIL-STD-202-301 | MIL-C-22520/5 |
| | MIL-STD-202-302 | ASTM B488 |

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2016-148)

Review activities:
Army - AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil .


Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:03Z
Check the source to verify that this is the current version before use.